ICT Strategy — Automated Bot Technical Specification

A precise, implementation-ready ruleset defining each ICT sub-signal algorithmically, and how to combine them into a single composite confirmation signal for automated trade execution.

---

1. Objective & Design Philosophy

The bot should NOT fire trades on any single ICT concept in isolation. Each concept below is turned into a boolean or scored sub-signal. A trade is only taken when a required combination of sub-signals aligns within a defined time window and price zone — this is the "confluence" or "confirmation stack" model. The spec below gives objective, codeable rules for every subjective ICT term so an AI/developer can implement it without guesswork.

2. Data & Timeframe Requirements

- **Timeframes needed:** HTF (H4 or D1) for bias/liquidity mapping, MTF (H1 or M15) for structure shifts, LTF (M1–M5) for precise entry timing.
- **Instrument scope:** Works best on liquid FX pairs, indices, or majors with clean session behavior (matches your existing 26-pair FX backtesting universe).
- **Required OHLCV data:** Open, High, Low, Close, Volume (or tick volume) per candle, timestamped in UTC, convertible to EST for session logic.
- **Swing point detection:** Use a fractal method — a swing high is a candle whose high is greater than N candles on each side (commonly N=2); a swing low is the inverse. This must run on HTF and MTF independently.

3. Algorithmic Definitions of Each ICT Component

3.1 Liquidity Pools

- **Buy-side liquidity (BSL):** the price level of the most recent unmitigated swing high (equal highs count as reinforced liquidity — group highs within 0.05%–0.1% of each other as one pool).
- **Sell-side liquidity (SSL):** the price level of the most recent unmitigated swing low, same equal-low grouping logic.
- **Liquidity grab / sweep confirmation:** price wicks through the BSL/SSL level by at least X pips (configurable, e.g. 2–5 pips on majors) and the candle closes back on the opposite side of that level within the same or next 1–3 candles on the entry timeframe.
- **Mitigation state:** once swept, mark that liquidity pool as "used" so the bot does not re-trigger on the same level.

3.2 Fair Value Gaps (FVG)

- **Bullish FVG:** 3 consecutive candles where candle 1's high < candle 3's low. The gap zone = [candle1.high, candle3.low].

• **Bearish FVG:** 3 consecutive candles where candle 1's low > candle 3's high. The gap zone = [candle3.high, candle1.low].

• **Minimum size filter:** discard FVGs smaller than a configurable threshold (e.g. 0.02% of price, or ATR(14)/10) to avoid noise on low timeframes.

• **Fill/rebalance state:** track each FVG's fill percentage as price re-enters the zone; mark "filled" once price has traded through ≥50% (or 100%, configurable) of the gap.

• **Validity window:** an FVG is only tradeable if unfilled and formed within the last N candles (configurable, e.g. last 20–50 candles on the entry timeframe) to keep it relevant.

3.3 Order Blocks (OB)

• **Bullish OB:** the last down-close candle immediately before a displacement move (see 3.6) that breaks structure upward. OB zone = [candle.low, candle.high] of that down candle (some implementations use only the candle body, not the wicks — pick one and keep it consistent).

• **Bearish OB:** the last up-close candle immediately before a displacement move that breaks structure downward.

• **Validation rule:** an order block is only "confirmed" if the subsequent displacement candle(s) produce a Market Structure Shift (3.5) — this filters out random candles.

• **Mitigation:** once price returns into the OB zone and reacts (reverses) at least once, mark as tested; if price closes fully through the OB, mark as invalidated.

3.4 Kill Zones (Session Time Filters)

Session	Time Window (EST)	UTC Equivalent
Asian Kill Zone	20:00 – 00:00	01:00 – 05:00
London Open Kill Zone	02:00 – 05:00	07:00 – 10:00
New York Open Kill Zone	07:00 – 10:00	12:00 – 15:00
London Close Kill Zone	10:00 – 12:00	15:00 – 17:00

Note: adjust UTC offsets for DST (EST vs EDT) — recompute the window daily using the exchange's actual local time, not a hardcoded UTC offset. The bot should only evaluate NEW trade signals when current time falls inside an active kill zone window; positions can still be managed outside it.

3.5 Market Structure Shift (MSS) / Break of Structure (BOS)

• **Bullish MSS:** price closes above the most recent confirmed lower-high (in a downtrend) or above the most recent swing high (in a range) on the MTF/LTF.

• **Bearish MSS:** price closes below the most recent confirmed higher-low (in an uptrend) or below the most recent swing low.

• **Distinction from BOS:** BOS = continuation break in the direction of the existing trend; MSS = the first break that signals a potential reversal. Track current trend state (uptrend/downtrend/range) via a simple higher-high/higher-low vs lower-high/lower-low state machine to classify which one just occurred.

3.6 Displacement

- A candle (or 2–3 consecutive candles) whose range is ≥ 1.5x–2x the average true range (ATR) of the prior N candles, AND whose body-to-wick ratio is high (e.g. body ≥ 60–70% of total range), AND which produces at least one Fair Value Gap.
- Displacement is the trigger that upgrades a normal swing candle into a valid Order Block (see 3.3).

3.7 Optimal Trade Entry (OTE) Zone

- After a confirmed MSS/displacement leg, draw a Fibonacci retracement from the start to the end of that leg.
- OTE zone = the 61.8%–79% retracement band of that leg.
- Entries are only valid if price retraces into this band AND that band overlaps with an unfilled FVG or unmitigated Order Block — this overlap is a key confluence requirement, not an either/or.

3.8 Premium / Discount Array

- Take the current dealing range = most recent significant swing high to swing low (HTF or MTF, configurable).
- Equilibrium (50%) splits the range. Below 50% = discount (buy zone), above 50% = premium (sell zone).
- Rule: only take long confirmations when price is in discount; only take short confirmations when price is in premium. This is a directional filter layered on top of everything else.

3.9 Power of Three (AMD) / Daily Bias Filter

- Accumulation phase: measured as the Asian session range (high/low) — store as daily reference range.
- Manipulation phase: detect as a liquidity sweep of the Asian range high or low occurring during the London Open Kill Zone.
- Distribution phase: the directional move expected during the New York session, opposite to the manipulation sweep direction (i.e., if Asian low was swept, bias = bullish distribution).
- Use this as a daily directional bias filter — the bot only takes trade signals aligned with the current AMD-implied bias for that day.

3.10 Judas Swing

- A sharp, low-follow-through move in the first 15–30 minutes of a session (typically London or NY open) that sweeps liquidity and then reverses.
- Detected as: a swing formed within the first N minutes of a kill zone, followed by an MSS in the opposite direction within the same kill zone window.

4. Composite Confirmation Signal Logic

Define a scoring system where each sub-signal contributes points. A trade is only triggered when the total score crosses a minimum threshold AND certain sub-signals are mandatory (not optional).

Sub-Signal	Mandatory?	Points
Active kill zone window (3.4)	Yes (hard filter)	0 (gatekeeper)
Daily AMD bias aligned (3.9)	Yes (hard filter)	0 (gatekeeper)
Liquidity sweep / grab (3.1)	Yes	2
Market Structure Shift in trade direction (3.5)	Yes	2
Displacement candle confirming MSS (3.6)	Yes	1
Price in Premium/Discount zone matching direction (3.8)	Yes (hard filter)	0 (gatekeeper)
Entry price inside OTE 62–79% zone (3.7)	No	1
Entry overlaps unfilled FVG (3.2)	No (either/or with OB)	1
Entry overlaps unmitigated Order Block (3.3)	No (either/or with FVG)	1
Judas Swing pattern present (3.10)	No (bonus)	1

- **Hard gatekeepers** (kill zone active, AMD bias aligned, correct premium/discount side) must ALL be true or the bot does not evaluate a trade at all — no scoring bypasses these.
- Of the remaining weighted signals, require sweep + MSS + displacement (5 points, all mandatory) as the core trigger.
- Require at least 1 additional point from OTE zone, FVG overlap, or OB overlap — i.e., price must be retracing into SOME discount/premium structural zone, not just any retracement.
- Judas Swing is a bonus/confidence signal only — useful for position sizing (larger size when present) rather than as a strict gate.

4.1 Confirmation State Machine (Pseudocode)

```
STATE: IDLE, WAITING_MSS, WAITING_ENTRY_ZONE, IN_TRADE

on new_candle(tf=LTF):
if not in_kill_zone(now): return
if not daily_bias_set(): compute_daily_bias() # from AMD logic 3.9

if state == IDLE:
if detect_liquidity_sweep(direction=daily_bias):
mark_sweep_point()
state = WAITING_MSS

elif state == WAITING_MSS:
if detect_mss(direction=daily_bias):
if is_displacement(mss_candle):
register_order_block(mss_candle)
register_fvg_if_any(mss_candle)
state = WAITING_ENTRY_ZONE
```

```
else:
state = IDLE # invalid, no displacement

elif state == WAITING_ENTRY_ZONE:
if price_in_discount_or_premium(direction=daily_bias) and (
price_in_ote_zone() or price_in_unfilled_fvg() or price_in_unmitigated_ob()
):
entry_price = current_price
stop_loss = sweep_extreme +/- buffer # beyond the liquidity grab wick
take_profit = next_opposing_liquidity_pool_or_fvg()
place_trade(entry_price, stop_loss, take_profit, direction=daily_bias)
state = IN_TRADE
elif zone_invalidated(): # price closed through OB/FVG without reacting
state = IDLE

elif state == IN_TRADE:
manage_trade() # trailing stop, partial TP, breakeven logic
```

5. Entry, Stop Loss & Take Profit Rules

- **Entry:** limit order placed at the near edge of the FVG/OB (not market order, to avoid slippage on the retracement).
- **Stop loss:** placed beyond the liquidity sweep extreme (the wick that grabbed liquidity), plus a buffer of 1–3 pips or 0.1×ATR to avoid getting stopped out by noise.
- **Take profit (primary):** the next opposing liquidity pool (equal highs/lows or unmitigated swing point) in the trade direction.
- **Take profit (secondary/partial):** nearest unfilled FVG or 1:2 risk-reward, whichever comes first — take partial profit here and move stop to breakeven.
- **Minimum risk-reward filter:** discard any setup where projected R:R < 1:2 based on the identified TP1 target.

6. Risk Management Layer

- Position size per trade: fixed fractional risk (e.g. 0.5–1% of account equity) based on stop-loss distance.
- Max concurrent trades: cap at 1–2 per correlated currency group to avoid compounding correlated risk (relevant given your 26-pair FX universe).
- Daily loss limit / health gate: pause new entries for the day if daily drawdown exceeds a set threshold (e.g. 2–3%) — reuse the health-gate concept from your Beast V2 engine.
- Session filter: no new trades outside defined kill zones, regardless of signal strength.
- News filter (optional but recommended): block entries within X minutes of high-impact calendar events for the traded instrument.

7. Backtesting & Validation Notes

- Backtest each sub-signal's standalone hit rate first (does liquidity-sweep-then-MSS actually precede favorable moves more than random?) before trusting the composite score.

• Walk-forward test across multiple sessions and pairs — ICT behavior is session- and instrument-dependent (works differently on indices vs FX majors vs exotics).

• Watch for look-ahead bias: FVGs and order blocks are only confirmed AFTER the displacement candle closes — make sure the backtest engine doesn't use future candle data to validate a zone that wouldn't have been known in real time.

• Log every gatekeeper/score breakdown per trade so you can later tune which components add real edge vs which just add noise — this lets you prune weak sub-signals (e.g. maybe Judas Swing detection isn't adding value and can be dropped).

8. Configuration Parameters Summary

Parameter	Suggested Default	Purpose
Swing fractal N	2	Swing high/low detection sensitivity
Equal high/low grouping tolerance	0.05%–0.1%	Merge nearby liquidity levels
Sweep confirmation buffer	2–5 pips	Confirm genuine liquidity grab
FVG minimum size	ATR(14)/10	Filter noise gaps
FVG validity window	20–50 candles	Discard stale gaps
Displacement ATR multiple	1.5x–2x	Qualify strong impulse candles
OTE zone	61.8%–79%	Entry retracement band
Min risk:reward	1:2	Trade quality filter
Risk per trade	0.5%–1%	Position sizing
Daily loss cap	2%–3%	Health-gate circuit breaker

This specification is designed to be handed to a developer or AI code-generation tool to implement as an automated strategy module. It is educational/technical documentation only, not financial advice — validate thoroughly via backtesting and forward-testing on a demo account before risking live capital.